Malek Badawi

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EDUCATION

Louisiana State University (LSU), Baton Rouge, LA **Expected May 2028**
Bachelor of Science in Computer Science (Software Engineering) *GPA: 4.17/4.30*
Minors: Mathematics and Honors Research
Honors: Stamps Scholar (full-ride national scholarship), President's Future Leaders in Research, President's Honor Roll
Coursework: LLM Application Development, Advanced Data Structures and Algorithms, Object-Oriented Programming
University College London – Artificial Intelligence Concepts & Applications, Summer 2025 (Distinction)

EXPERIENCE

Undergraduate Researcher **March 2026 – Present**
Pennington Biomedical Research Center – Imaging Department *Baton Rouge, LA*
- Evaluated the lab's BRISQUE MRI QC pipeline against manual review, identified its failure to capture MRI-specific artifacts, and supported the transition to radiologist-validated MRIQC.
- Consolidated FreeSurfer outputs for ~460 subjects into a unified symlink-based directory and deployed MRIQC in an Apptainer container on a shared Linux/NFS server.
- Built Bash and MATLAB batch-processing workflows across seven concurrent studies with 500+ subjects, adding validation safeguards and resolving file-processing, naming, and variable-scope errors.

Software & Data Engineering Intern **September 2025 – June 2026**
Automotive Data Analytics *Remote*
- Built a three-endpoint SMS authentication service using JavaScript, Azure Logic Apps, Azure Table Storage, and Twilio, supporting OTP verification, code expiration, and session validation.
- Built Microsoft Fabric ETL pipelines consolidating 10,000+ customer records from multiple dealership systems into a centralized lakehouse, automating refreshes and supporting Power BI reporting.

PROJECTS

Community Coffee Onboarding Platform | *React, Python, FastAPI, SQL, GitHub Actions*
- Developing a full-stack employee onboarding platform for Community Coffee through LSU's Applied Intelligence Lab, translating onboarding requirements into structured, role-specific workflows.
- Building React interfaces, FastAPI backend services, and SQL-based data models for managing employee roles, onboarding tasks, and workflow relationships across the application.
- Implementing automated testing and CI with GitHub Actions while designing the platform architecture to support future AI-assisted role and task mapping.

PFT Navigator | *Unreal Engine, Blueprints, UMG, JSON, Perforce, Figma*
- Developing an interactive navigation platform for LSU's Patrick F. Taylor Hall, connecting room, course, faculty, and department search results to locations within an existing 3D digital twin.
- Building Unreal Engine navigation features with Blueprints and UMG, including destination selection, room highlighting, floor-path visualization, and guided routing through the building model.
- Integrating structured JSON data with digital-twin assets and collaborating in a six-person team using Perforce for version control and Figma for interface design.

SKILLS

Languages: Java, Python, TypeScript, JavaScript, C, SQL
Frameworks & Libraries: React, Next.js, Node.js, Vitest, OpenCV, MediaPipe, NumPy
Tools & Platforms: Git/GitHub, GitHub Actions, Linux/Unix, REST APIs, Postman, SQLite, Azure, Microsoft Fabric, Power BI, Twilio, Vercel, Apptainer/Singularity
Research Computing: FreeSurfer, MRIQC, pybids, BIDS, NIfTI, DICOM
Spoken: Arabic (fluent), English (fluent), Spanish (conversational)

LEADERSHIP

Student Government Senator **August 2025 – Present**
Louisiana State University – Baton Rouge, LA
- Elected by the College of Engineering; serve on the Student Auxiliary Services and Academic Affairs Committees, contributing to initiatives involving campus technology resources and ethical AI policy.